

Seeing Inside by Pushing from Outside: The Story of Mechanics Based Tomography

A Long-Form Essay on the Body of Work by Yue Mei and Colleagues

Let me take you on a journey through one of the more elegant ideas to emerge from the intersection of solid mechanics, inverse problems, and biomedical imaging over the past decade. The idea is called Mechanics Based Tomography — MBT for short — and while it is still relatively young as a research program, the intellectual arc it traces is both deep and surprisingly wide, touching everything from the mathematical structure of ill-posed optimization to the clinical dream of detecting tumors without ever sending a wave through the body. The principal architect of this program is Yue Mei, who began this work during his doctoral training at Texas A&M University under Sevan Goenezen and has since extended it as a faculty member at Dalian University of Technology in China, collaborating with an expanding network of researchers across continents — from Stéphane Avril at Mines Saint-Étienne in France to Corey Neu and Deva Chan in the United States.

To understand what MBT is trying to do, you have to first understand what came before it, and why those earlier approaches left an important gap.

The Landscape Before MBT: Elastography and Its Limits

The broader field in which MBT sits is called elastography — the imaging of mechanical properties, typically stiffness, inside biological tissues. The fundamental clinical motivation is straightforward: many diseases alter the mechanical properties of tissue. Cancerous tumors, for instance, are often significantly stiffer than the surrounding healthy tissue. If you could create a spatial map of stiffness throughout a region of the body, you would have a diagnostic tool that captures something fundamentally different from what conventional imaging — ultrasound, MRI, CT — provides. Those modalities image density, water content, chemical composition. Elastography images mechanical behavior.

The earliest and most intuitive version of elastography is something every physician already does: palpation. You press on tissue, and if you feel something hard beneath the surface, that is diagnostic information. But palpation is qualitative, limited to shallow depths, and entirely dependent on the clinician's experience. Elastography seeks to formalize and extend that principle.

The mainstream approach to elastography, which has been developed since the early 1990s, involves inducing some kind of mechanical excitation inside the tissue — usually through ultrasound-based radiation force or through an external vibrator — and then measuring the resulting deformation field using ultrasound imaging or MRI. The measured deformation field is then used, together with the governing equations of solid mechanics, to solve an inverse problem: given how the tissue deformed, what must its stiffness distribution be?

This approach has been enormously productive. Magnetic resonance elastography, or MRE, is now FDA-approved for liver fibrosis staging. Ultrasound-based shear wave elastography is widely available in clinical practice. But there is a crucial limitation embedded in this paradigm. To solve the inverse problem, you need interior displacement measurements — you need to know how points deep inside the tissue moved. This requires either MRI, which gives you full three-dimensional displacement fields but is expensive, slow, and not always accessible, or ultrasound, which gives you axial displacement along the beam direction but introduces significant noise and is limited in its ability to capture the full tensorial nature of deformation.

Here is where MBT enters the picture with a genuinely different proposition.

The Core Idea: What If You Only Had Surface Measurements?

The central question that Yue Mei and Sevan Goenezen asked, beginning around 2015 and 2016, was this: can you reconstruct the interior stiffness distribution of a heterogeneous solid — say, a piece of tissue with a tumor inside — using only measurements taken on the surface?

Think about what this means physically. You have a solid object. You press on it from the outside, and you measure how its surface deforms using cameras — specifically, a digital image correlation system, which is a well-established optical technique for measuring full-field surface displacements with high precision. You also measure the forces you applied. And from that information alone — surface displacements and boundary forces — you try to figure out whether there is something stiffer or softer hiding inside, where it is, how big it is, and how stiff it is relative to the background.

This is, on its face, an outrageously ambitious proposition. You are trying to infer three-dimensional interior properties from two-dimensional surface data. The information has to travel, so to speak, from the boundary inward through the governing physics — the equations of equilibrium, the constitutive law relating stress to strain. And the inverse problem is severely ill-posed, meaning that many different interior stiffness distributions could, in principle, produce very similar surface displacement patterns, especially in the presence of measurement noise.

But Mei and Goenezen had a key insight that made progress possible. They realized that by applying multiple independent loading configurations and measuring the resulting surface displacements for each, they could dramatically increase the information content available for the inversion. If you push from the left and measure the surface, then push from the top and measure again, and so on, each loading scenario interrogates the interior from a different mechanical vantage point. The combination constrains the inverse problem much more tightly than any single loading.

This is, in a deep sense, tomographic — hence the name Mechanics Based Tomography. Just as CT scanning reconstructs interior density by sending X-rays through the body from many angles, MBT reconstructs interior stiffness by mechanically loading the body from many directions and observing the resulting surface deformations.

The First Feasibility Studies: 2016–2017

The earliest published work laying the foundation for MBT appeared in 2016, when Mei and colleagues published a paper in the *International Journal of Solids and Structures* on estimating the nonhomogeneous elastic modulus distribution from surface deformations. This paper established the computational framework: a constrained optimization problem where the unknown is the spatially varying shear modulus field, the forward model is a finite element solution of the elasticity equations, and the objective function measures the mismatch between the predicted and measured surface displacements.

The optimization is driven by an adjoint method, which is an efficient way to compute the gradient of the cost function with respect to thousands of unknown shear modulus values simultaneously. The adjoint approach avoids the need to compute sensitivities one variable at a time, which would be computationally prohibitive when the unknown field is discretized on a fine finite element mesh.

In 2017, the formal introduction of the MBT name came in a paper published in *Sensors*, titled “Mechanics Based Tomography: A Preliminary Feasibility Study,” authored by Mei, Sicheng Wang, Xin Shen, Stephen Rabke, and Goenezen. This paper presented a

systematic numerical study using simulated experiments. The team created synthetic displacement data by solving forward problems with known stiffness distributions — backgrounds containing one or two stiff inclusions — and then added noise to mimic the imperfections of real experimental measurements. They then ran their inverse algorithm using only the noisy boundary displacements and force measurements.

The results were remarkably encouraging. The algorithm recovered the locations, approximate shapes, and stiffness contrasts of the inclusions without making any prior assumptions about how many inclusions existed, where they were, or what shape they had. The shear modulus was treated as a continuous, freely varying field — every node in the finite element mesh was an independent unknown. When multiple loading cases were used, the reconstructions improved substantially, particularly at lower noise levels. The paper also examined the effect of having displacement data on only part of the boundary, simulating a more realistic experimental scenario where cameras might not capture the entire surface.

At the same time, a companion paper by Mei, Tajderi, and Goenezen in the International Journal of Applied Mechanics addressed a subtlety that is easy to overlook but critically important: what happens when you already know something about the elastic modulus in part of the domain? This paper on regularizing biomechanical maps for partially known material properties introduced total variation diminishing and total contrast diminishing regularization strategies. These are not just mathematical niceties. In a clinical setting, you might know from palpation or from another imaging modality that the tissue near the surface has normal stiffness. Incorporating that partial knowledge properly can dramatically improve the reconstruction in the interior, where the problem is most ill-conditioned.

Tackling the Hard Problems: Boundary Sensitivity, Size Dependence, and Regularization

Between 2016 and 2019, Mei and collaborators systematically identified and addressed a series of fundamental challenges in the MBT framework. One of the most important was boundary sensitivity — the observation that the quality of the stiffness reconstruction degrades for inclusions located near the boundary where displacement data is available, and particularly for inclusions near boundaries where no displacement data is available.

The 2016 paper in the Journal of Applied Mechanics, by Mei, Kuznetsov, and Goenezen, introduced a clever modification to the cost function: weighting the displacement mismatch by a function that depends on the strain field. The idea is that regions of high

strain are more informative about local stiffness, so giving them more weight in the optimization improves the contrast of the reconstruction. In a simplified one-dimensional coupled model, they derived an analytical expression showing why this weighting works, and demonstrated improved results in two-dimensional simulations.

A related but distinct issue emerged in a 2019 paper published in *Applied Sciences*: the solution to the inverse problem using the standard L2 norm cost function is sensitive to the sizes of the inclusions. Smaller inclusions tend to be recovered with lower stiffness contrast than larger ones, even if their true stiffness is the same. Mei, Sun, and Zhao showed that this is not a numerical artifact but is inherent to the regularization. They proposed a size-dependent cost function that weights the contribution of each inclusion region according to its area, effectively compensating for the bias. Simulated experiments showed that this approach improves shear modulus contrast recovery and reduces size sensitivity.

These are the kinds of papers that do not get a lot of attention outside the specialist community, but they represent the careful, unglamorous work of understanding exactly why an algorithm succeeds or fails, and then engineering targeted fixes. The intellectual style is characteristic of the MBT research program as a whole: a mix of rigorous analysis, often including analytical one-dimensional models that build intuition, combined with systematic computational studies.

The Constitutive Question: Linear vs. Nonlinear Elasticity

One of the most practically important questions the MBT group addressed was whether a linear elastic model is adequate for the inverse problem when the tissue is undergoing large deformations and the actual material behavior is nonlinear. This matters because biological soft tissues — the primary targets of MBT — are famously nonlinear. They stiffen dramatically at large strains, and their stress-strain curves are exponential, not linear.

In a 2018 paper in the *International Journal of Mechanical Sciences*, Mei, Stover, Kazerooni, Srinivasa, Hajhashemkhani, Hematiyan, and Goenezen presented a comparative study using both linear elastic and neo-Hookean hyperelastic constitutive models. They used experimental displacement data from a silicone composite sample — a physical phantom containing two stiff inclusions of different sizes — subjected to uniaxial extension. The displacements were measured using digital image correlation.

The finding was nuanced and important. Using a linear elastic model on a sample undergoing large deformations overestimates the stiffness contrast between the inclusions and the background. But the linear model still correctly identifies the

locations and relative stiffnesses of the inclusions. Moreover, the linear model is significantly cheaper to solve — no iterative Newton-Raphson steps, no concerns about convergence. A one-dimensional analytical model helped explain why: the linear elastic inverse solution effectively extrapolates the tangent stiffness from the current strain state, and because the material stiffens with strain, this tangent stiffness is higher than the small-strain shear modulus.

The practical implication is that for applications where you primarily need to detect and localize heterogeneities — which is the main clinical question in tumor detection — the linear elastic model may be entirely sufficient. When you need quantitatively accurate stiffness values, particularly at large strains, the hyperelastic model becomes necessary.

Experimental Validation: The 2021 Milestone

The papers described so far relied primarily on synthetic data — displacement fields generated by solving forward problems with known stiffness distributions, then corrupted with noise. The critical question of whether MBT works with real experimental data was addressed definitively in a 2021 paper in the *Journal of the Mechanics and Physics of Solids* by Goenezen, Kim, Kotecha, Luo, and Hematiyan.

This paper validated MBT using surface displacement measurements obtained from digital image correlation on a silicone-based composite specimen. The specimen was a cube with embedded stiffer inclusions, and the surface displacements were measured using regular camera resolution — not specialized high-speed or high-resolution cameras, but the kind of setup that would be accessible in many engineering labs.

The validation involved two parts. First, synthetic verification with the same geometry and material properties as the experimental specimen, to establish that the inverse algorithm works correctly in the noise-free case. Second, application to the actual experimental data, including all the messiness of real measurements — noise, imperfect boundary conditions, slight material nonuniformities.

The experimental results were encouraging. The algorithm successfully recovered the interior stiffness distribution, identifying the locations and approximate stiffness contrasts of the inclusions from surface-only measurements. The paper also introduced an additional innovation: a method for identifying the unknown traction distributions on the specimen boundaries. In real experiments, you know the total applied force from load cells, but the exact spatial distribution of tractions on the surface is not known. Mei and colleagues developed an adjoint-based optimization to solve for these tractions simultaneously with the stiffness distribution.

This traction identification capability is itself a significant contribution. Knowing the spatial distribution of tractions at an interface is important in many engineering contexts — from understanding load transfer in joints to characterizing contact mechanics — and the fact that it can be recovered from surface displacement measurements alone opens interesting possibilities beyond the original MBT application.

The Explicit Inverse Approach: Moving Morphable Inclusions

By 2020 and 2021, Mei had moved to Dalian University of Technology and began collaborating with Xu Guo and others on a fundamentally different strategy for solving the inverse problem. The standard approach — what might be called the implicit approach — treats the shear modulus at every finite element node as an independent unknown. This is maximally flexible, making no assumptions about the number, shape, or stiffness of inclusions. But it leads to a very high-dimensional optimization problem with thousands or tens of thousands of unknowns, which is expensive and potentially ill-conditioned.

The new idea, published in the *Journal of Applied Mechanics* in 2021, was to parameterize the inclusions explicitly using moving morphable inclusions, or MMIs. Instead of optimizing the stiffness at every node, you describe each inclusion using a small number of geometric parameters — its center coordinates, dimensions, orientation, and stiffness value — plus the contour of the inclusion described by B-spline curves. The total number of optimization variables drops from thousands to perhaps a few dozen. The price you pay is that you need to specify in advance how many inclusions to look for, and you assume that the stiffness distribution consists of a small number of discrete regions with uniform properties — an assumption that is reasonable for many medical and engineering applications.

The advantages of this explicit approach are substantial. First, the optimization converges much faster because the search space is so much smaller. Second, the reconstruction naturally produces sharp, well-defined inclusion boundaries rather than the blurred gradients typical of pixel-based reconstructions with regularization. Third, and perhaps most surprisingly, the method works well even without regularization at moderate noise levels, because the low-dimensional parameterization itself acts as an implicit regularizer.

This explicit approach represents a significant intellectual pivot for the MBT program. It moves the framework from a general-purpose tool for mapping arbitrary stiffness distributions to a more targeted tool for detecting and characterizing discrete features

— which is, arguably, the most clinically relevant application.

Branching Into New Domains: The Virtual Fields Method

In parallel with the development of MBT, Mei has also made significant contributions to a related but distinct inverse method: the Virtual Fields Method, or VFM. The VFM is a technique originally developed by Pierron and Grédiac for identifying constitutive parameters from full-field displacement measurements. Its key advantage over optimization-based methods like MBT is computational efficiency — the VFM converts the identification problem into a direct algebraic system that can be solved without iteration, provided the virtual fields are chosen appropriately.

Mei's contributions to the VFM have been substantial and complementary to the MBT work. In a 2019 paper in the *International Journal of Solids and Structures*, he and Stéphane Avril addressed the accuracy of shear modulus identification in incompressible elasticity using the VFM. They showed that the choice of virtual fields significantly affects the accuracy of the reconstruction, and they proposed improved strategies for selecting these fields.

In 2021, Mei, Deng, Guo, Zimmerman, Nguyen, and Avril published a major paper in the *Journal of Elasticity* presenting a general finite element framework for the VFM in nonlinear elasticity. This was a significant extension because the VFM had previously been developed mainly for linear problems, and extending it to nonlinear constitutive models — essential for biological soft tissues — required addressing new theoretical challenges around the selection of virtual fields in the presence of geometric and material nonlinearity.

A subsequent paper in *Computational Mechanics* introduced regularization into the VFM, combining the computational efficiency of the direct method with the robustness of regularized optimization. This hybrid approach proved effective on both numerical and experimental datasets.

The VFM and MBT work are deeply interrelated but address different practical scenarios. MBT is designed for situations where you only have surface measurements — no interior displacement data. The VFM is designed for situations where you have full-field interior measurements, typically from MRI or high-resolution DIC on thin specimens, and want to recover the stiffness distribution efficiently. Together, they represent a comprehensive toolkit for inverse problems in elasticity.

Clinical Frontiers: Cornea and Intervertebral Disc

The most recent and clinically exciting developments in Mei's research have extended the inverse problem framework to two specific anatomical targets: the cornea and the intervertebral disc.

The corneal work, published in *Computers in Biology and Medicine* in 2023, applied digital image correlation to the high-speed videos captured by the Corvis ST tonometer — a device already used clinically to measure intraocular pressure. The Corvis ST works by firing a puff of air at the cornea and recording the resulting deformation at high speed. Mei and colleagues showed that by applying DIC to these videos, you can extract full-field strain and strain rate maps of the cornea, not just the global deformation parameters that the device normally provides. Comparing normal and keratoconus corneas, they found that keratoconus corneas exhibit significantly larger shear strain and shear strain rate than normal corneas. This is important because keratoconus — a progressive thinning and bulging of the cornea — is one of the conditions where early biomechanical diagnosis could have the most clinical impact, particularly for screening patients before refractive surgery.

A subsequent study went further, combining incremental DIC analysis of Corvis ST images with machine learning to distinguish forme fruste keratoconus — a subclinical form of keratoconus — from normal corneas. The machine learning models using biomechanical DIC data outperformed the standard Corvis ST parameters, demonstrating the diagnostic value of the richer biomechanical information that the inverse problem framework provides.

The intervertebral disc work is perhaps the most ambitious clinical application to date. In a 2025 paper published in the *Proceedings of the Royal Society A*, Mei, Jin, Kang, Zhao, Chan, Neu, and Avril presented MRI elastography of the human intervertebral disc using the explicit inverse approach. The intervertebral disc is a layered structure with a soft, gel-like nucleus pulposus in the center surrounded by a stiffer, fibrous annulus fibrosus. Degeneration of the disc — a major cause of low back pain — alters the mechanical properties of both regions, and being able to map those properties in vivo could transform how we diagnose and track disc disease.

The explicit inverse method is particularly well-suited to this problem because the disc has a known anatomical structure — you know there is a nucleus and an annulus, and you know roughly where the boundary between them is. The optimization therefore focuses on just a few parameters: the shear moduli of the nucleus and annulus, and the geometric description of their interface. Using MRI-derived displacement fields from discs under compression and bending, the method yielded annulus-to-nucleus shear modulus ratios in the range of six to fifteen — consistent with the expected range from

independent mechanical testing.

A complementary study, also involving Mei as a co-author, appeared as a bioRxiv preprint in 2023 from a team led by Deva Chan at Purdue, combining MRI elastography with relaxometry measurements. This study demonstrated that strain imaging and stiffness estimation can serve as imaging biomarkers to distinguish healthy and degenerated discs, and that these mechanical measures provide information complementary to the relaxation-time-based measures that are more commonly used in clinical MRI.

The Broader Intellectual Contribution

Stepping back from the specific papers and results, there is a larger story about what Yue Mei and his collaborators have accomplished over the past decade. They have taken an inverse problem that many in the field would have dismissed as hopelessly ill-conditioned — reconstructing interior stiffness from surface-only measurements — and systematically demonstrated its feasibility through a combination of mathematical analysis, computational innovation, and experimental validation.

The progression of the work is instructive. It began with basic feasibility studies using synthetic data, moved to careful analysis of the mathematical pathologies — boundary sensitivity, size dependence, regularization artifacts — then proceeded to experimental validation with physical phantoms, and is now reaching into genuine clinical applications with real tissue data. At each stage, the work has been characterized by a combination of computational studies and analytical insight, often using simplified one-dimensional models to build physical understanding of why the algorithms succeed or fail.

The intellectual bridge between MBT, the VFM contributions, and the clinical applications is the deep learning and explicit parameterization thread. By moving from pixel-based implicit reconstructions to explicit geometric parameterizations — whether through moving morphable inclusions, B-spline interfaces, or topology description functions — Mei's group has found ways to make the inverse problem dramatically more tractable while retaining the essential physics. This is not a trivial observation. Many researchers in the inverse problems community are pursuing purely data-driven approaches using neural networks. Mei's work shows that there is enormous value in maintaining the physics-based framework while using structural assumptions to reduce the dimensionality of the unknown space.

For those of us in the broader soft tissue biomechanics community, the MBT program is worth watching for several reasons. First, the surface-only measurement paradigm is uniquely suited to applications where interior imaging is impractical or unavailable —

think of intraoperative assessment, point-of-care diagnostics, or even smartphone-based measurements where you can photograph the surface deformation of skin but cannot access an MRI scanner. Second, the mathematical and computational tools developed for MBT — adjoint-based optimization, regularization strategies for ill-posed problems, explicit geometric parameterizations — are transferable to a wide range of inverse problems in mechanics, including the kind of boundary-displacement-based elasticity reconstruction that several labs, including those working on projects like PAT-Scan, are pursuing.

Finally, there is something aesthetically satisfying about the MBT idea. It takes the oldest and most intuitive form of medical diagnosis — pressing on tissue and feeling what is inside — and elevates it to a rigorous, quantitative imaging modality by adding cameras, computers, and the full machinery of continuum mechanics. The stethoscope amplifies hearing. The microscope amplifies vision. MBT, in a sense, amplifies touch. And it does so by leaning not on new physics or new sensors, but on the power of the inverse problem — the idea that if you understand the governing equations well enough, you can extract information that seems, at first glance, to be inaccessible.

Key Papers in the MBT Research Arc

For those who wish to trace the full development, here is a chronological guide to the essential publications:

Mei, Kuznetsov, and Goenezen (2016). "Reduced Boundary Sensitivity and Improved Contrast of the Regularized Inverse Problem Solution in Elasticity." *Journal of Applied Mechanics*. This paper introduced the strain-weighted cost function and first articulated the concept that would become MBT.

Mei, Fulmer, Raja, Wang, and Goenezen (2016). "Estimating the Non-homogeneous Elastic Modulus Distribution from Surface Deformations." *International Journal of Solids and Structures*. The foundational computational framework.

Mei, Wang, Shen, Rabke, and Goenezen (2017). "Mechanics Based Tomography: A Preliminary Feasibility Study." *Sensors*. The paper that named and formally introduced MBT.

Mei, Tajderi, and Goenezen (2017). "Regularizing Biomechanical Maps for Partially Known Material Properties." *International Journal of Applied Mechanics*.

Luo, Mei, Kotecha, et al. (2018). "Characterization of the Stiffness Distribution in Two and Three Dimensions Using Boundary Deformations." *MRS Communications*. Extension to three-dimensional reconstruction.

Mei, Stover, Kazerooni, et al. (2018). "A Comparative Study of Two Constitutive Models Within an Inverse Approach." *International Journal of Mechanical Sciences*. Linear versus nonlinear constitutive models.

Zhao, Sun, and Mei (2019). "A Size-Dependent Cost Function to Solve the Inverse Elasticity Problem." *Applied Sciences*.

Mei and Avril (2019). "On Improving the Accuracy of Nonhomogeneous Shear Modulus Identification in Incompressible Elasticity Using the Virtual Fields Method." *International Journal of Solids and Structures*.

Liu, Sun, Deng, and Zhao (2020). "A Comparative Study of Direct and Iterative Inversion Approaches." *International Journal of Applied Mechanics*.

Mei, Du, Zhao, Zhang, Liu, and Guo (2021). "Moving Morphable Inclusion Approach: An Explicit Framework to Solve Inverse Problem in Elasticity." *Journal of Applied Mechanics*. The paradigm-shifting explicit approach.

Goenezen, Kim, Kotecha, Luo, and Hematiyan (2021). "Mechanics Based Tomography: Validation Using Experimental Data." *Journal of the Mechanics and Physics of Solids*. The experimental validation milestone.

Mei, Liu, Guo, Zimmerman, Nguyen, and Avril (2021). "General Finite-Element Framework of the Virtual Fields Method in Nonlinear Elasticity." *Journal of Elasticity*.

Wang, Yang, Cheng, Wang, and Mei (2023). "In-vivo High-speed Biomechanical Imaging of the Cornea Using Corvis ST and Digital Image Correlation." *Computers in Biology and Medicine*.

Davis, Gossett, Wilson, Kim, Mei, et al. (2023). "Intervertebral Disc Elastography to Relate Shear Modulus and Relaxometry." *bioRxiv preprint*.

Mei, Zhao, Xiao, Sun, et al. (2024). "Nondestructively Identifying the Mechanical Behavior of Soft Tissues Using Surface Deformation with an Explicit Inverse Approach." *Applied Mathematical Modelling*.

Mei, Jin, Kang, Zhao, Chan, Neu, and Avril (2025). "MRI Elastography of Human Intervertebral Disc Based on an Explicit Inverse Approach." *Proceedings of the Royal Society A*.

What you see in this body of work is not just a collection of papers but the deliberate construction of a research program — each paper addressing a specific gap, building on what came before, and opening the way for what comes next. The trajectory from synthetic feasibility to experimental validation to clinical application is clear and

intentional. And the underlying intellectual commitment — that the physics of solid mechanics, properly interrogated, contains far more information than naive measurement alone can extract — remains as compelling today as when the first MBT paper was published nearly a decade ago.